

Sensitivity Analysis of Promotional Strategies of Mobile Telephone Service Providers II

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ABSTRACT

Even though there are several companies that provide mobile telephone services, two companies, namely Airtel and Hutch, control the student segment in Karnataka. In order to attract customers, both companies offer several special promotional schemes, month to month. Some of these promotional schemes are: unlimited free SMS, unlimited free outgoing calls within home network, and full talk-time. The effect of these promotional schemes has been large-scale brand-switching behaviour of customers.

This study extends the authors' earlier analysis of the extent of brand-switching in the mobile telephone services among students in Bangalore, Karnataka. The Markov brand-switching model is extended to include brand loyalty, and the effect of each combination of promotional strategies adopted by the two companies is analysed in this new model. The optimal mixes of promotional strategies for the two companies are then found using game-theoretic techniques. Finally, as the analysis above is based on sample data, sensitivity analysis is performed to assess the effect of sampling/estimation error on the results.

Key words: Promotional strategies, Brand-switching behaviour, Markov brand-switching model, Game theoretic techniques, Sensitivity analysis

1. INTRODUCTION

The Markov brand-switching model is a relatively simple application of the theory of stochastic processes to consumer behaviour. In modeling brand-switching behaviour, it makes two simplifying assumptions.

The first assumption, called the Markov property, is that customers “forget” the past history of brands used, and only consider the current brand they are using when deciding whether to switch brands or not. Thus, at any point of time, the brand-switching behaviour of customers is not determined by which brands they have used in the past, the order in which they used past brands, and their past experiences with these brands. The only data relevant to their brand-switching decision is the brand they are currently using.

The second assumption is that one-period transition probabilities between states are constant at all points of time – i.e. the percentages of customers switching between brands in the next period are fixed. Thus, if there are only two brands, say A (Airtel) and H (Hutch), the one-period transition probabilities are given by p (the percentage of customers currently using brand A who switch over to brand H in the next period) and q (the percentage of customers currently using brand H who switch over to brand A in the next period). These one-period transition probabilities are represented in a one-period transition probability matrix as follows:

$$\begin{array}{cc} & \text{Next period state} \\ & \begin{array}{cc} & A & H \end{array} \\ \text{Current state} & \begin{array}{c} A \\ H \end{array} \left(\begin{array}{cc} 1-p & p \\ q & 1-q \end{array} \right) \end{array}$$

The one-period transition probability matrix is used to describe the evolution of the system over time. Thus, if $\pi_{A,t}$ is the percentage of customers using brand A at time t , and $\pi_{H,t}$ is the percentage of

customers using brand H at time t , then the percentages of customers using the brands A and H at time $t + 1$ is given by the matrix equation $\pi_{t+1} = \pi_t \cdot P$, where $\pi_t = (\pi_{A,t} \quad \pi_{H,t})$ is the state vector at time t .

As the system described above evolves, eventually a “dynamic equilibrium” is achieved whereby the switching between brands does not alter the distribution between the brands. This is achieved when $p \cdot \pi_A = q \cdot \pi_H$, where π_A and π_H are the percentages of customers using brands A and H, respectively. Such a condition is called “steady-state.” The steady-state distribution of the customers under the above conditions would then be given by

$$\pi_A = \frac{q}{p+q} \text{ and } \pi_H = \frac{p}{p+q}.$$

The steady-state distribution of the customers between the brands thus represents their long-run market shares, under the two assumptions above.

The game theoretic framework presents a simple method for evaluating and determining decisions under conditions of conflict. The simplest form of game is that of a “2x2 two-player zero-sum game,” in which there are only two “players” or participants, each of them having to choose between only two possible alternative courses of action, in which the outcome is determined by the alternatives chosen by both of the players, and in which the “payoff” resulting to one player is exactly the reverse of that resulting to the other player. Such a game can be represented by a payoff matrix (conventionally representing payoffs to the row player):

		Player H	
		T ₁	T ₂
Player A	S ₁	a	b
	S ₂	c	d

where a is the payoff to player A when player A chooses strategy S₁ and player H chooses strategy T₁, b is the payoff to player A when player A chooses strategy S₂ and player H chooses strategy T₁, c is the payoff to player A when player A chooses strategy S₂ and player H chooses strategy T₂, and d is the payoff to player A when player A chooses strategy S₂ and player H chooses strategy T₂. The payoffs to player H are exactly the negatives of these.

Assuming that these are the only alternative courses of action available to the players, two possibilities may result. The first possibility occurs if one of the elements a, b, c, or d of the payoff matrix is simultaneously the least element in the row in which it lies and the largest element in the column in which it lies. Such an element is called a saddlepoint element. If a saddlepoint exists in the payoff matrix, then the strategies corresponding to the row and the column in which the saddlepoint lie are the optimal strategies for the row and column players, respectively. The strategies are optimal in the sense that, assuming that the other player does not change their strategy, neither player can increase their payoff by changing their strategy.

The second possibility occurs if no saddlepoint exists in the payoff matrix. In this case, no single strategy is optimal in the above sense for either player. Instead, both players must “optimally mix” the two strategies available to them, shifting between the two strategies in a fixed proportion. These “optimal mixing” proportions are given by

$$p_1 = \frac{d - c}{(a + d) - (b + c)} \text{ and } p_2 = \frac{a - b}{(a + d) - (b + c)} \text{ for player A,}$$

where p_1 represents the percentage of the time player A should use strategy S₁ and p_2 represents the percentage of time of time player A should use strategy S₂, and

$$q_1 = \frac{d - b}{(a + d) - (b + c)} \text{ and } q_2 = \frac{a - c}{(a + d) - (b + c)} \text{ for player H,}$$

where q_1 represents the percentage of the time player B should use strategy T₁ and q_2 represents the percentage of time of time player A should use strategy T₂. The “expected value” of the game, i.e. the average payoff to player A is given by $E_V = \frac{a \cdot d - b \cdot c}{(a + d) - (b + c)}$. The average payoff to player H is just the negative of this quantity.

Several studies have used the Markov model to study brand-switching behaviour (e.g. Bass et al, 1984). Pfeifer and Carraway (2000) used Markov chains to model the switching of customers between different levels of recency of buying with the firm. In the context of valuing customer relationships, Rust et al (2001)

suggested the use of a Markov process to develop a customer equity model that takes into account brand switching. Libai et al (2002) have also suggested the application of the Markov model between customer segments to estimate customer equity. On the other hand, Sun et al (2003) have studied the impact of promotions of brand-switching, but their approach was based on logit choice models, and they proposed the use of dynamical structural models for better results.

The present study is an extension of the authors' paper (2006*), incorporating brand loyalty into the Markov brand-switching model. Thus, if the initial market shares of Airtel (A) and Hutch (H) are denoted by m_A^o, m_H^o , and the proportions of brand-loyal customers of Airtel and Hutch are denoted by θ_A, θ_H , then the long-run market shares may be expressed as

$$\pi_A^L = \theta_A \cdot m_A^o + \frac{q}{p+q} ((1-\theta_A) \cdot m_A^o + (1-\theta_H) \cdot m_H^o)$$

$$\pi_H^L = \theta_H \cdot m_H^o + \frac{p}{p+q} ((1-\theta_H) \cdot m_H^o + (1-\theta_A) \cdot m_A^o)$$

The steady-state distribution in the extended model is thus further dependent upon two additional sets of parameters: the initial distribution and the proportions of brand-loyal customers. The question arises as to whether these initial conditions would affect the promotional mix of the mobile telephone service providers.

2. DATA AND METHODOLOGY

The objective of the present study is to extend the Markov model by introducing brand loyalty, and to assess the impact of brand loyalty on the "optimal promotion mix." As with the authors' earlier study, the scope of the present study was limited to studying brand-switching behaviour between Airtel and Hutch among students in Bangalore. In particular, the study was designed to investigate the effect of promotional strategies on brand-switching, so as to determine the "optimal promotion mix" that Airtel and Hutch should offer for the student segment. The different promotional strategies considered for the study were: unlimited free SMS (denoted by S_1 for Airtel and T_1 for Hutch), unlimited free outgoing calls within home network (denoted by S_2 for Airtel and T_2 for Hutch), and full talk time (denoted by S_3 for Airtel and T_3 for Hutch).

The data was collected from a random sample of 100 students in Bangalore who were Airtel users and an independent random sample of 100 students in Bangalore who were Hutch users. Each respondent was asked whether they would switch service provider under each strategy combination employed by Airtel and Hutch. For each strategy combination the sample switching rates were computed, and the long-run market shares of Airtel and Hutch were computed using these sample switching rates in the extended Markov model, under different initial conditions. These long-run market shares were then considered as the payoffs in a 3x3 two-player, zero-sum game (after adjustment by deducting 50% from them). The optimal promotional strategy mix was then computed using game theoretic techniques. To analyse the robustness of the optimal strategy mix, sensitivity analysis was performed, using the 95% confidence limits in the switching rates. Finally, the results of the analysis under different initial conditions were compared to assess the impact of the initial conditions on the "optimal promotion mix."

3. ANALYSIS AND INTERPRETATION

The sample estimates one-period switching rates between the brands under each strategy combination are given in Table 1 below:

Table 1: Sample estimates of one-period switching rates between Airtel and Hutch under each strategy combination

Percentages switching from Hutch to Airtel				Percentages switching from Airtel to Hutch				
	T ₁	T ₂	T ₃		S ₁	S ₂	S ₃	
S ₁	62%	43%	61%		S ₁	17%	18%	13%
S ₂	42%	66%	69%		S ₂	55%	10%	09%
S ₃	28%	26%	42%		S ₃	70%	63%	11%

The resulting payoff matrix for Airtel is given in Table 2 below:

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Table 2: Payoff matrix for Airtel (base case)

		Hutch		
		T ₁	T ₂	T ₃
Airtel	S ₁	28.48%	20.49%	32.43%
	S ₂	-6.70%	36.84%	38.46%
	S ₃	-21.43%	-20.79%	29.25%

Strategies S₃ and T₃ are clearly dominated strategies, and the optimal mixing percentages for S₁ & S₂ for Airtel and T₁ & T₂ for Hutch are given by: p₁ = 84.49%, p₂ = 15.51%, and q₁ = 31.73%, q₂ = 68.27%. The expected value for Airtel is Ev = 23.025%. This means that, in the base case, the optimal promotion mix for Airtel is to offer unlimited free SMS 84.49% of the time and unlimited free outgoing calls 15.51% of the time, alternatingly, while the optimal promotion mix for Hutch is to offer unlimited free SMS 31.73% of the time and unlimited free outgoing calls 68.27% of the time, alternatingly, and subsequently the expected outcome would be a 23.025% shift in the long-run market share in favour of Airtel.

As the data used in the computations above was sample data, sensitivity analysis was performed to analyse the robustness of the above optimal strategies. In performing the sensitivity analysis, 95% confidence limits were used. In the “optimistic case” (from Airtel’s point of view), the lower 95% confidence limit (i.e. $\hat{p} - 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$) was used to estimate the switching rate from Airtel to Hutch, and the upper 95% confidence limit (i.e. $\hat{q} + 1.96\sqrt{\frac{\hat{q}(1-\hat{q})}{n}}$) was used to estimate the switching rate from Hutch to Airtel. In the “pessimistic case” (from Airtel’s point of view), the upper 95% confidence limit (i.e. $\hat{p} + 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$) was used to estimate the switching rate from Airtel to Hutch, and the lower 95% confidence limit (i.e. $\hat{q} - 1.96\sqrt{\frac{\hat{q}(1-\hat{q})}{n}}$) was used to estimate the switching rate from Hutch to Airtel.

The payoff matrix for Airtel (in the optimistic case) is given in Table 3 below:

Table 3: Payoff matrix for Airtel (optimistic case)

		Hutch		
		T ₁	T ₂	T ₃
Airtel	S ₁	38.12%	33.43%	41.67%
	S ₂	3.31%	44.81%	45.84%
	S ₃	-12.38%	-10.75%	41.39%

Again, strategies S₃ and T₃ are clearly dominated strategies, and the optimal mixing percentages for S₁ & S₂ for Airtel and T₁ & T₂ for Hutch are in this case: p₁ = 89.95%, p₂ = 10.15%, and q₁ = 24.64%, q₂ = 75.36%. The expected value for Airtel in this case is Ev = 34.585%. This means that, in the optimistic case (i.e. if all conditions are favourable for Airtel), the optimal promotion mix for Airtel is to offer unlimited free SMS 89.95% of the time and unlimited free outgoing calls 10.15% of the time, alternatingly, while the optimal promotion mix for Hutch is to offer unlimited free SMS 24.64% of the time and unlimited free outgoing calls 75.36% of the time, alternatingly, and subsequently the expected outcome would be a 34.585% shift in the long-run market share in favour of Airtel.

The payoff matrix for Airtel (in the pessimistic case) is given in Table 4 below:

Table 4: Payoff matrix for Airtel (pessimistic case)

		Hutch		
		T ₁	T ₂	T ₃
Airtel	S ₁	18.30%	6.60%	22.42%
	S ₂	-16.70%	28.13%	30.40%
	S ₃	-30.45%	-30.63%	15.37%

Again, strategies S_3 and T_3 are clearly dominated strategies, and the optimal mixing percentages for S_1 & S_2 for Airtel and T_1 & T_2 for Hutch are in this case: $p_1 = 79.30\%$, $p_2 = 20.70\%$, and $q_1 = 38.08\%$, $q_2 = 61.92\%$. The expected value for Airtel in this case is $Ev = 11.055\%$. This means that, in the pessimistic case (i.e. if all conditions are unfavourable for Airtel), the optimal promotion mix for Airtel is to offer unlimited free SMS 79.30% of the time and unlimited free outgoing calls 20.70% of the time, alternately, while the optimal promotion mix for Hutch is to offer unlimited free SMS 38.08% of the time and unlimited free outgoing calls 61.92% of the time, alternately, and subsequently the expected outcome would be a 11.055% shift in the long-run market share in favour of Airtel.

The results of the analysis using the extended Markov model show that the optimal promotion mix is not affected by either the initial market share or the proportions of brand-loyal customers. However, there is an effect of the initial conditions in terms of the shift in market share. Table 5 below shows the effect of varying levels of brand loyalty and varying market shares.

Table 5: Results for different levels of brand loyalty and different market shares

		$m_A^o = 60\%, m_H^o = 40\%$			$m_A^o = 50\%, m_H^o = 50\%$			$m_A^o = 40\%, m_H^o = 60\%$		
θ_A	θ_H	base	opt	pess	base	opt	pess	base	opt	pess
5%	0%	23.84%	35.05%	12.22%	23.70%	34.97%	12.03%	23.57%	34.89%	11.83%
0%	5%	21.57%	32.84%	9.83%	21.20%	32.47%	9.53%	20.84%	32.05%	9.22%
5%	5%	22.38%	33.36%	11.00%	21.88%	32.86%	10.50%	21.38%	32.36%	10.00%
10%	5%	23.18%	33.82%	12.17%	22.55%	33.24%	11.48%	21.91%	32.66%	10.78%
5%	10%	20.91%	31.66%	9.78%	20.05%	30.74%	8.98%	19.18%	29.82%	8.17%
10%	10%	21.72%	32.13%	10.95%	20.72%	31.13%	9.95%	19.72%	30.13%	8.95%
15%	10%	22.53%	32.59%	12.12%	21.40%	31.51%	10.92%	20.26%	30.43%	9.73%
10%	15%	20.26%	30.43%	9.73%	18.90%	29.01%	8.42%	17.53%	27.59%	7.12%
15%	15%	21.07%	30.90%	10.90%	19.57%	29.40%	9.40%	18.07%	27.90%	7.90%
20%	15%	21.88%	31.36%	12.07%	20.25%	29.78%	10.37%	18.61%	28.20%	8.68%
15%	20%	19.61%	29.20%	9.68%	17.75%	27.28%	7.87%	15.88%	25.36%	6.07%
20%	20%	20.42%	29.67%	10.84%	18.42%	27.67%	8.84%	16.42%	25.67%	6.84%
25%	20%	21.23%	30.13%	12.01%	19.01%	28.05%	9.82%	16.96%	25.98%	7.62%
20%	25%	18.96%	27.98%	9.62%	16.60%	25.55%	7.32%	14.32%	23.13%	5.01%
25%	25%	19.77%	28.44%	10.79%	17.27%	25.94%	8.29%	14.77%	23.44%	5.79%

The results show about 5% variation in the shift in long-run market share between different initial market shares; the variation increases with higher levels of brand loyalty. The impact of Airtel's brand loyalty level on the shift in long-run market share decreases by 2-4% with decreasing initial market share, while the impact of Hutch's brand loyalty level on the shift in long-run market share increases by 6-8%.

4. DISCUSSION AND CONCLUSION

On the basis of the above analysis, it follows that Airtel, by alternately using the strategies of unlimited free SMS 84.49% of the time (on average) and unlimited free outgoing calls within home network 15.51% of the time (on average), can achieve on average a 23.025% shift in the long-run market share in its favour, and that the best response of Hutch would be to also offer unlimited free SMS and unlimited free outgoing calls within home network alternately, but in the proportion 31.73% and 68.27% (on average), respectively. Moreover, the sensitivity analysis shows a difference of more than 10% between the optimistic case outcomes and the pessimistic case outcomes, indicating a high degree of variation. The initial market share and level of brand loyalty was not found to affect the optimal promotion mix, but does affect the shift in long-run market share. The impact of brand loyalty is higher at lower levels of initial market share. More accurate estimates should be obtained in order to be able to calculate the mixing proportions accurately, and to assess the final impact of the promotional mix.

In a sense, these results concur with the practices of service providers. Airtel and Hutch do offer the promotional schemes described above, and vary the offers available month-to-month. The analysis above suggests that this should be done in a systematic, calculated manner. A possible approach would be to estimate market shares, levels of brand loyalty, and switching rates periodically, and to update the optimal

promotion mix with current data. In this way, the estimates would be more accurate, and any changes could be detected and incorporated into the analysis. This would result in a more dynamic, reactive strategy.

There are several limitations in the present study, relating to the assumptions of the Markov model and the methodological simplifications employed.

To begin with, the Markov property is a very restrictive assumption. Consumer behaviour, and in particular brand-switching, involves several factors, especially the history of experiences with brands. The extended model incorporates brand loyalty, but does not completely resolve the limitations of the Markov property. At best the assumption of the Markov property could be justified by arguing that brand-switching behaviour is determined mainly by brand differentiation, marginalizing the role of past experiences with brands.

Another weakness in the Markov model is the assumption of stationarity. In reality, switching rates themselves would depend on several factors, and would change over time. At best these switching rates could be taken to be constant for the short-run, in which case a long-run analysis would be misleading. The long-run market shares could be interpreted as “implied” market shares, and could therefore still function as objectives.

Another limitation arises from the restriction of scope only to brand-switching behaviour between Airtel and Hutch, ignoring switching to and from other service providers. The preceding analysis can be generalized to more than two service providers. Similarly, the analysis can also be generalized to include more promotional strategies, and more synthetic promotional strategies. Some strategies may be redundant or inferior, as was the case with the third strategy considered, viz. full talk time. A more detailed study of what factors affect brand choice and brand-switching behaviour in students would enable a more realistic model of the same.

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